



Space System Engineering and
Operations

*Attitude\orbit determination
& Control*

A.A.2024/25

AOCS – Attitude, Orbit Control Subsystem - when the orbit guidance is not performed on board

Functional chain of a satellite which encompasses:

- **attitude** and **orbit** sensors
- **attitude estimation** and **guidance**
- **attitude** and **orbit control** algorithms
- **attitude** and **orbit** control actuators

It might include an **orbit estimation** function usually called **Navigation**

GNC - Guidance Navigation and Control - commonly used for the on-board segment, when the satellite position is controlled in closed loop

Functions in charge of:

- **targeted orbit** and **attitude** computation
- **attitude** and **orbit determination**
- **attitude** and **orbit control**

Attitude Determination and Control Subsystem (ADCS) is a **sub-set of the AOCS**

Functional requirements

The AOCS shall provide the hardware and software capabilities and performances:

- to perform the ***attitude measurement, estimation, guidance and control*** needed for the mission;
- to perform the ***orbit control*** manoeuvres, as specified by the mission requirements;
- to ***ensure a safe state*** of the spacecraft at any time, ***including emergency and anomaly situations***, according to failure management requirements;
- to ensure the mission availability, as specified at satellite level.

Acquisition & keeping reqs

The AOCS shall provide **during all phases** of the mission the capability to **acquire and keep all attitudes** necessary to perform the mission.

Attitude determination reqs

The AOCS shall provide, as specified by the mission requirements, the hardware and software means for **autonomous on-board attitude determination**.

Orbit determination

If a navigation function is necessary for the mission, the AOCS shall provide the hardware and software means for autonomous **on-board determination of the spacecraft orbital state** which includes position, velocity and time.

Mission pointing requirements

The AOCS shall ensure that the **attitude guidance and pointing** specified by the mission requirements, during the mission operational phase are met.

Orbital acquisition and maintenance reqs

The AOCS shall provide the capability for **achieving orbit control manoeuvres** specified by mission analysis.

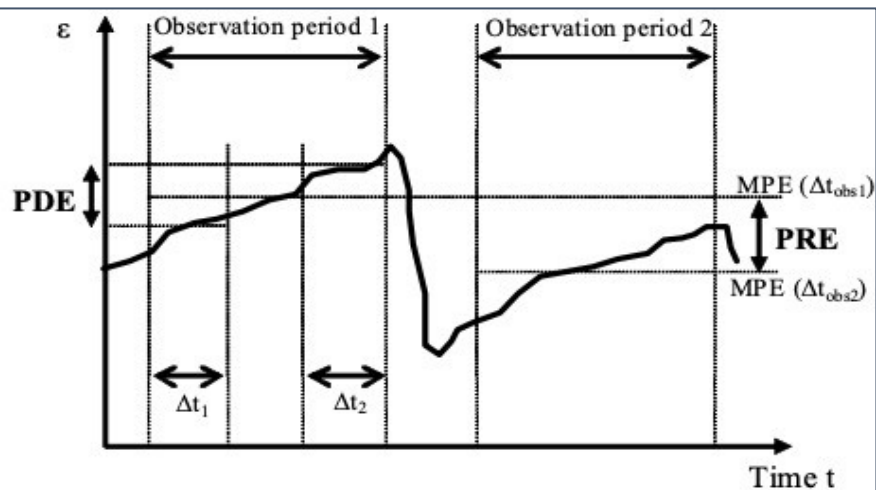
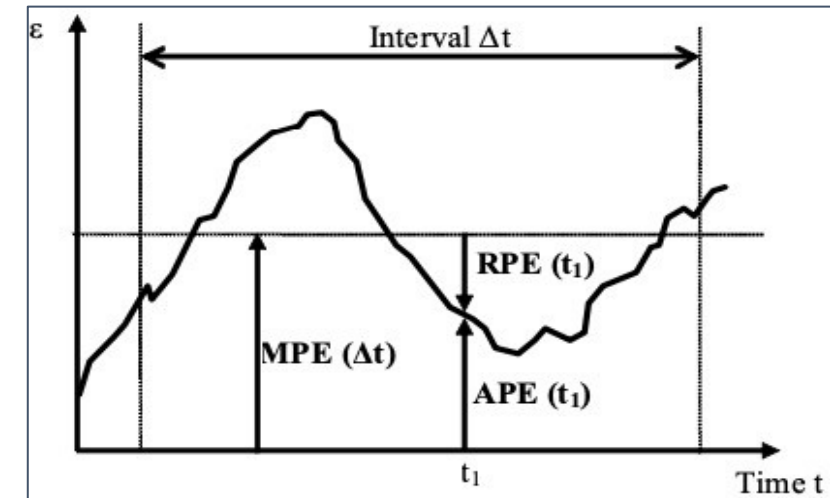
Safe mode

In case of major anomaly, the AOCS shall provide the autonomous capability to **reach and control safe pointing attitude** and angular rates to ensure the integrity of the spacecraft vital functions, including power, thermal and communications.

- The entry into safe mode shall be commandable by ground TC.
- The return from safe mode shall be commandable by ground TC

Index	Name	Definition
AKE	Absolute Knowledge Error	Difference between the actual parameter (attitude, geolocation, etc.) and the known (measured or estimated) parameter in a specified reference frame.
APE	Absolute Performance Error	Difference between the target (commanded) parameter (attitude, geolocation, etc.) and the actual parameter in a specified reference frame.
MKE	Mean Knowledge Error	Mean value of the AKE over a specified time interval Δt .
MPE	Mean Performance Error	Mean value of the APE over a specified time interval Δt .
RKE	Relative Knowledge Error	Difference between the APE at a given time within a time interval, Δt , and the MKE over the same time interval.
RPE	Relative Performance Error	Difference between the APE at a given time within a time interval, Δt , and the MPE over the same time interval.

AOCS performance indexes



Index	Name	Definition
KDE	Knowledge Drift Error	Difference between MKEs taken over two time intervals separated by a specified time, Δt_s , within a single observation period.
PDE	Performance Drift Error	Difference between MPEs taken over two time intervals separated by a specified time, Δt_s , within a single observation period.
KRE	Knowledge Reproducibility Error	Difference between MKEs taken over two time intervals separated by a specified time, Δt_s , within different observation periods.
PRE	Performance Reproducibility Error	Difference between MPEs taken over two time intervals separated by a specified time, Δt_s , within different observation periods.

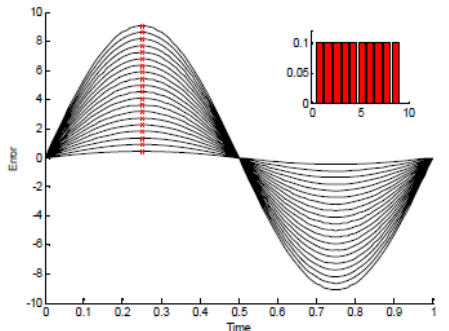
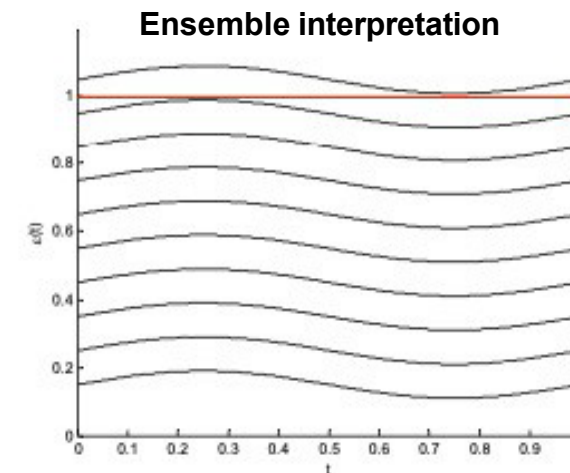
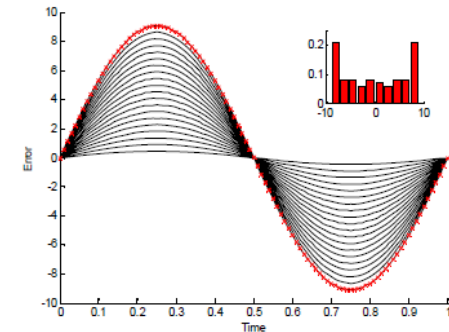
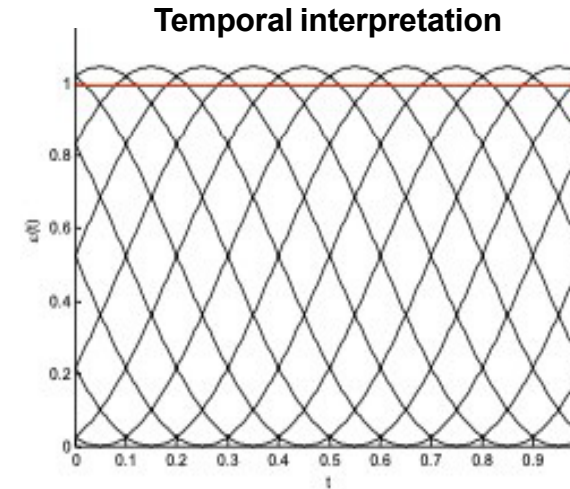
AOCS Performance Requirements – statistical interpretation

Assume that $APE < 1$ deg, at 90% probability:

- **90% of the time** for all members of the ensemble (**temporal interpretation**),
- 100% of the time for **90% of the ensemble** (**ensemble interpretation**),
- Intermediate case (**mixed interpretation**)

The **statistical interpretation** of the requirement must be considered according to mission specific needs

- 99.7% (general case) or if applicable 3σ (Gaussian distribution)
- 95.5% (general case) or if applicable 2σ (Gaussian distribution)
- 68.3% (general case) or if applicable 1σ (Gaussian distribution).



Absolute Pointing requirements – accuracy in control

The AOCS shall ensure during the operational mission phase an absolute pointing performance of TBS microradians, at TBS % confidence level, using the TBS (temporal, ensemble or mixed) statistical interpretation

Attitude Absolute Knowledge requirements - accuracy in determination

The AOCS shall ensure during the operational phase of the mission an on-board absolute attitude knowledge performance of TBS microradians, at TBS % confidence level, using the TBS (temporal, ensemble or mixed) statistical interpretation

Relative Pointing requirements

The AOCS shall ensure, during the operational phase of the mission, an on-board relative pointing performance of TBS microradians over a duration of TBS seconds, at TBS % confidence level, using the TBS statistical interpretation

Orbit Knowledge requirements - accuracy in determination

The navigation function shall provide the on-board orbit estimation with an accuracy of TBS metres (for position), TBS metres per second (for velocity) and TBS seconds (for time), in a TBS unambiguous space and time reference frame, at TBS % confidence level, using the TBS (temporal, ensemble or mixed) statistical interpretation

Orbit control requirements - accuracy in control

The AOCS shall perform the Delta-V commanded by the ground for the orbit control with an accuracy better than:

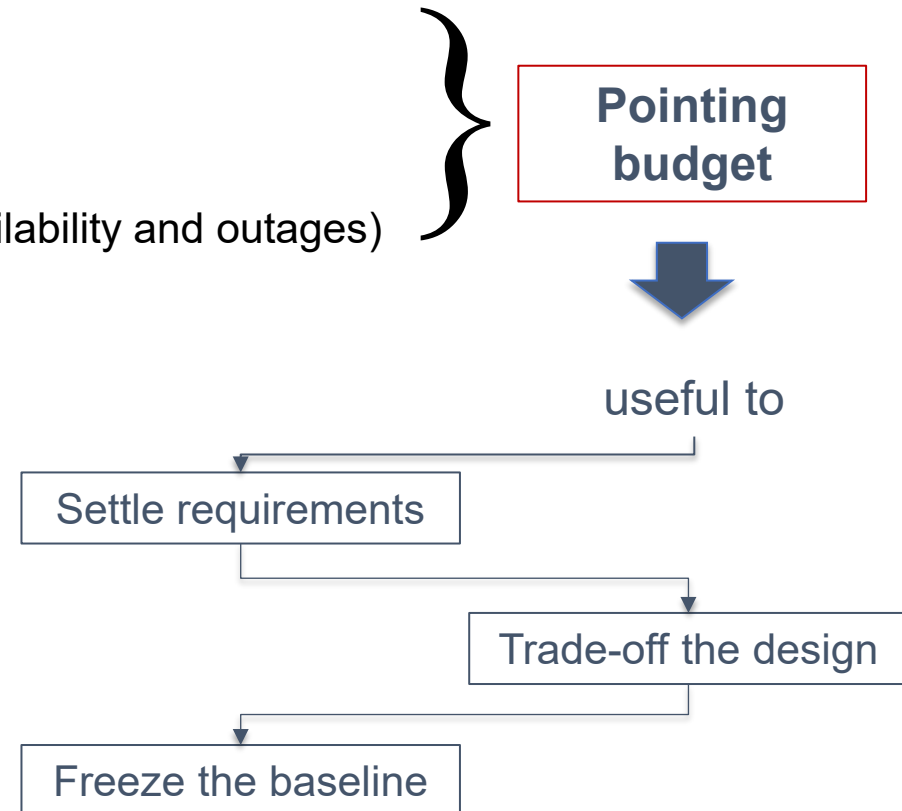
1. TBS % of the Delta-V magnitude along the commanded direction.
2. TBS % of the Delta-V magnitude on the perpendicular directions.

TBS= to be specified

AOCS Budgets

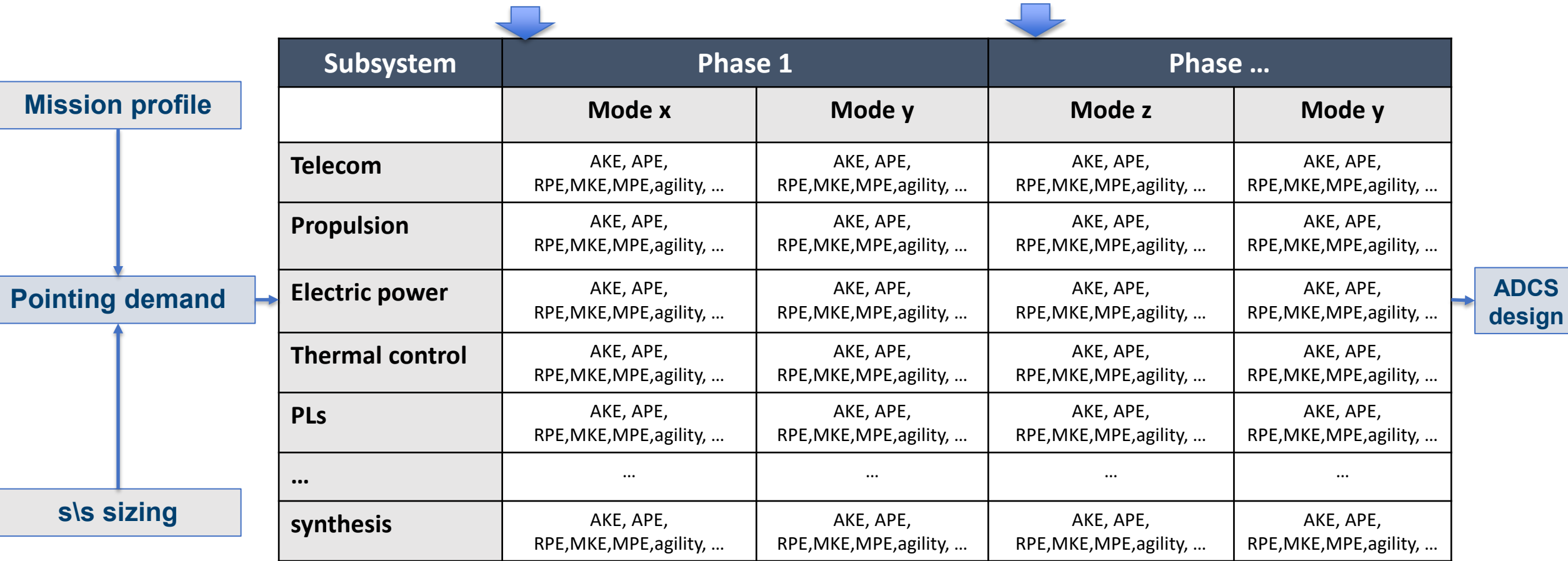
The AOCS shall **provide budgets** for the following performances:

- absolute attitude pointing budgets, APE
- on-board absolute attitude knowledge budgets, AKE
- relative attitude pointing budgets, RPE
- duration budgets (mode transitions, agility, convergence, AOCS availability and outages)
- contribution to propulsion related budgets,
- orbit correction performance budgets



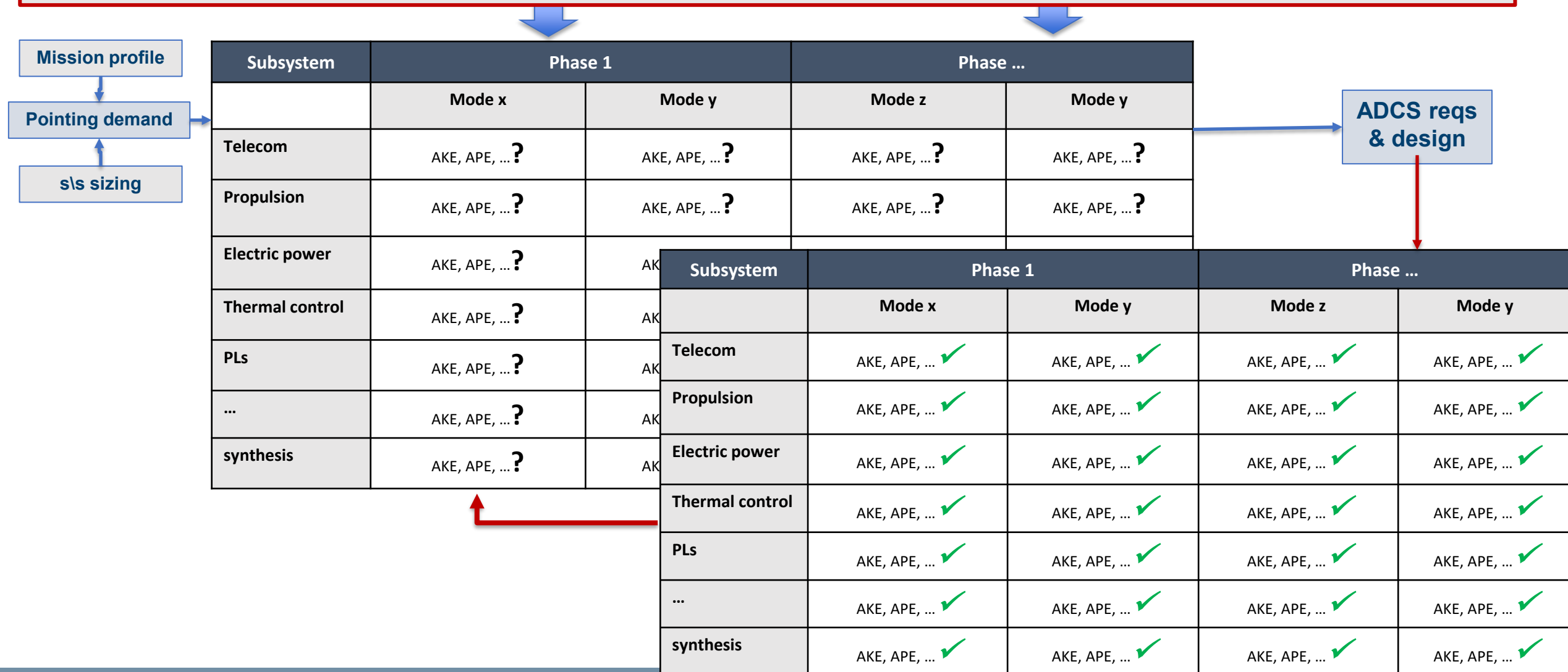
AOCS Design process: pointing budget as input

Fill the table with pointing knowledge\control **needed** by each s/s in each operational mode



AOCS Design process: pointing budget as input

Fill the table with ADCS SIZED FOR each s/s in each operational mode and check the consistency



AOCS Design process

1. Define control modes
2. Summarize the mission pointing **direction** requirements by control mode:
 - dependent on:
 - the p/l (e.g. Nadir pointing, scanning, inertial target, etc)
 - The mission phase\operations
 - The specific onboard s/s
3. Summarize p/l and mission pointing knowledge\control **accuracy** requirements
 - Dependent on:
 - Vehicle tasks, phases

Note: pointing accuracy typical values: SA $\rightarrow 4^{\circ}$ - 10° ; HGA $\rightarrow 0.1^{\circ}$ - 0.5° ; optics and cameras $\rightarrow 0.001^{\circ}$ - 0.1°
4. Define the **maneuvering** required for the mission including:
 - Angles, duration, rates and frequency

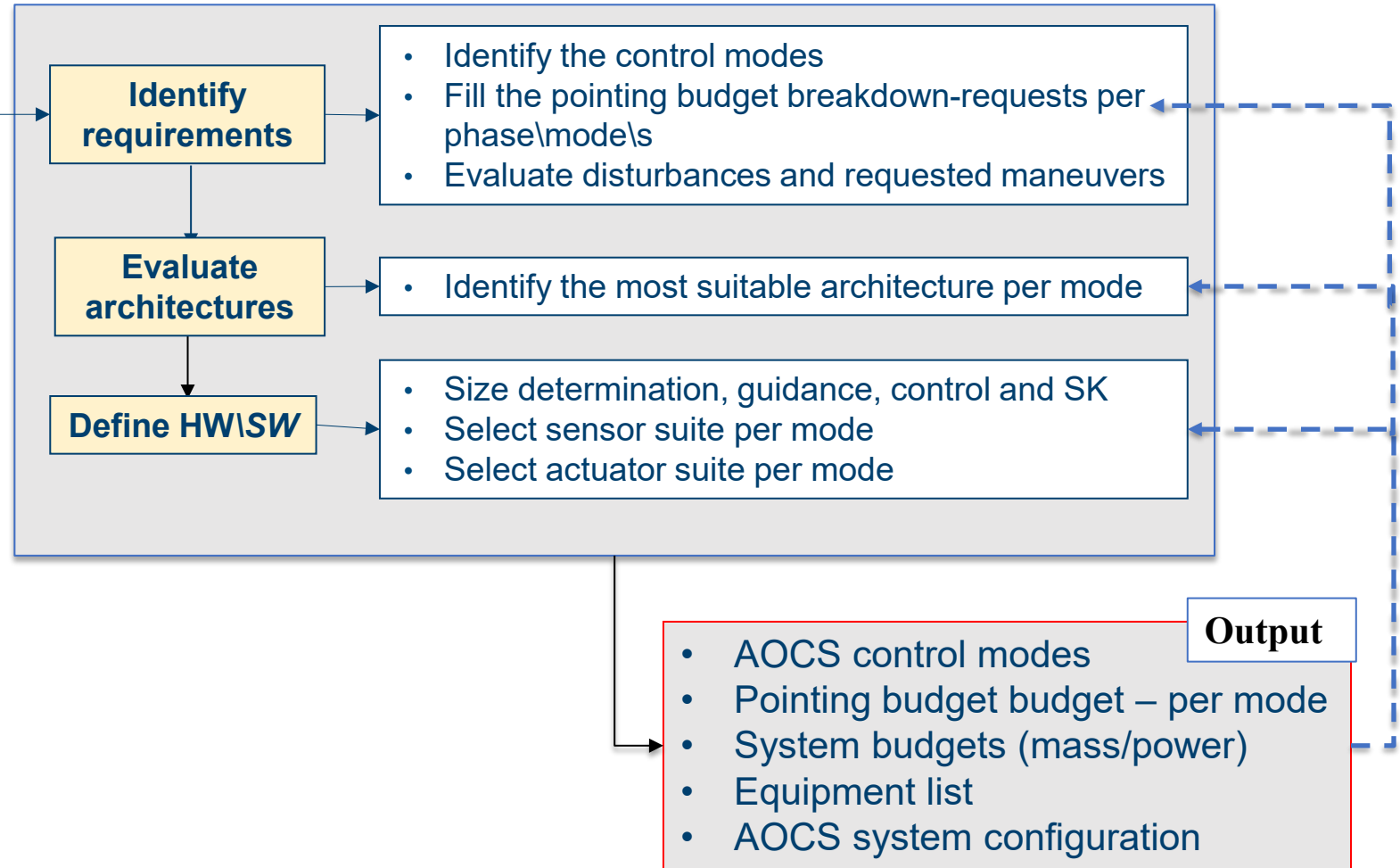
AOCS Design process

5. **Quantify** the internal and external **disturbance torques** (e.g. perturbations, sloshing, flexible appendages)
6. **Select the ADCS architecture**. That selection asks for knowledge of:
 - p/l pointing \ thermal\ power reqs
 - thrust vector control reqs
 - control authority
 - admissible attitude errors for translational Δv maneuvers
 - maneuver slew rates reqs/constraints (a driver if greater than $0.5^\circ/\text{s}$)
 - approximate solar panel area $\rightarrow$ if wings are greater than the s/c area $\rightarrow$ 3-axis stabilized preferred
7. **Select and size the major hardware components** (sensors/actuators)
8. Define the attitude determination and control **algorithms**
9. Conduct possible trade – off studies to improve the reqs and baseline selections
10. **Test the HWSW at incremental level of complexity and fidelity** (e.g. VM, EM, FM)

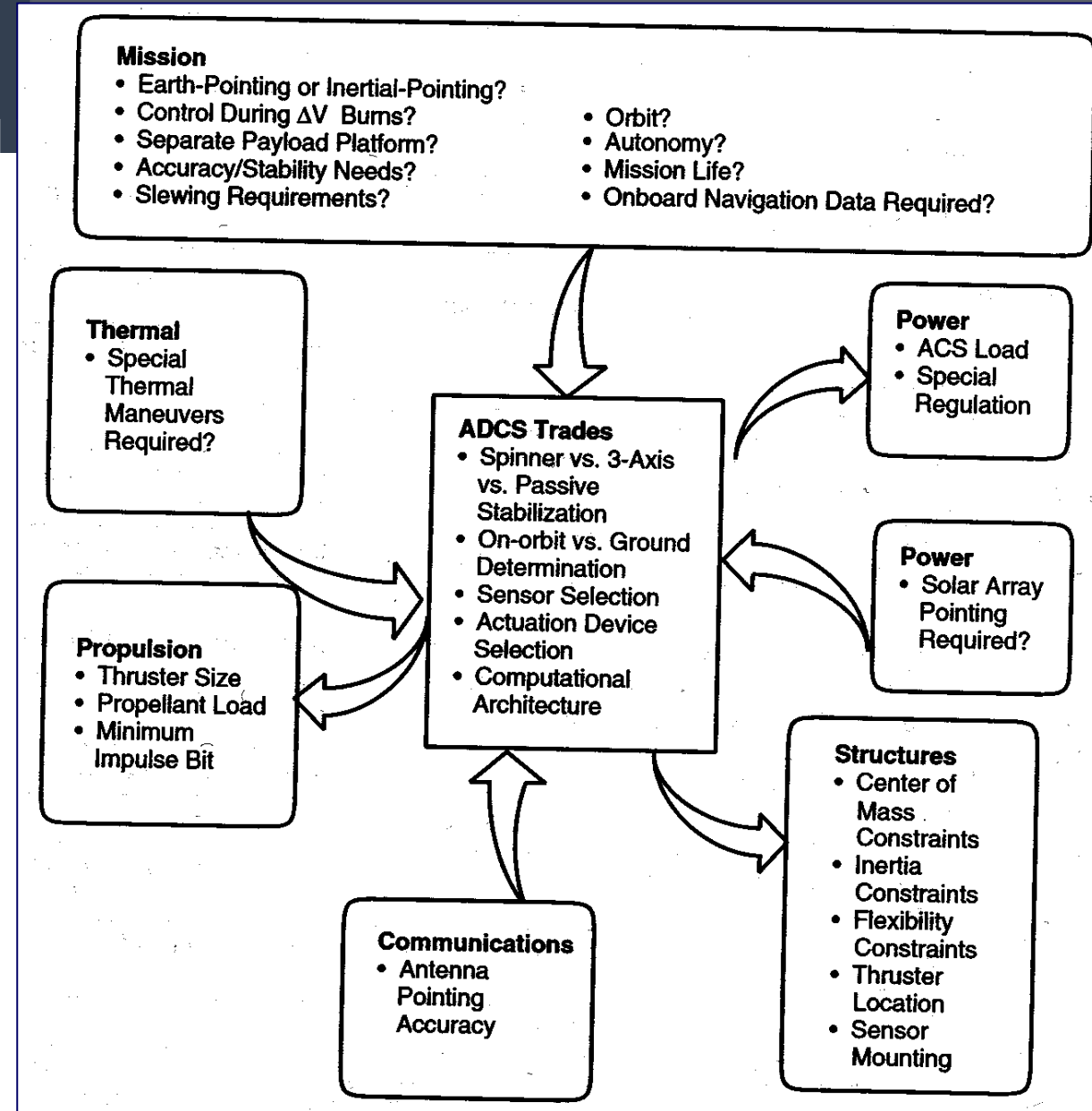
AOCS Design Process synthesis

Information needed

- Mission Profile-lifetime
- Spacecraft configuration
- Phase\modes
- Pointing budget (as request)
- Power budget
- Thermal environment



How other S\S influence the ADCS reqs and trade offs



ADCS requirements sources: examples

<i>Requirement from payload</i>	<i>Information needed</i>
element to be pointed	Entire p/l or subset (radiators, antennas,etc)
Pointing direction	Reference definition
Pointing range	All of the possible pointing directions
Pointing knowledge	Pointing direction knowledge in real-time or after the pointing
Pointing accuracy	Max rate of change of angular orientation
Slew rate	Reorientation from a point to another in a given time
Exclusion zones	i.e. far from the Sun

<i>Other Requirements</i>	<i>Information needed</i>
Sun pointing	May be needed for power generation/thermal control, even for p/l
Pointing while thrusting	May be needed for guidance corrections
Communications antenna pointing	Toward a ground station or relay satellite
Safe mode	Safe mode definition

1) Control Modes

Control modes are defined taking into account:

- **Flexibility** (i.e. capacity to adapt and solve more circumstances, without ambiguities, with simple and effective configuration options)
- **Autonomy** (i.e. capacity to be operative without the need of ground intervention)
- **Redundancy** (i.e. capacity to be operative in case of the loss of some HW components)
- **Performance** (i.e. capacity to provide control with various control and computing performances)

Example

AOCS Mode ex	Tasks	Characteristics
Safe Hold Mode	• Detumble the satellite and attitude determination • Point the solar panels towards the Sun	Most reliable HW shall be used Minimization of power consumption Rough pointing acceptable Completely autonomous
Standby Mode	• Point the solar panels towards the Sun • Point antennas towards Ground Station • Minimize drag during non-operative phases • Point a radiator to the sky for thermal dissipation • Other (depending on mission, ...)	More complex HW should be involved Better pointing than in Safe, could be worst than in Operative mode Typically autonomous Different GNC functions from Safe
Operational Mode	• Fulfill the main target of the mission (image acquisition, antenna pointing,...)	Most precise HW involved Most precise pointing achieved, even implying complex filtering or control techniques Typically Ground-assisted
Orbit Control Mode	• Perform orbital manoeuvres	Propulsion involved Thrust parasitic torques to be controlled Typically Ground-assisted
Transfer Mode	• Control the attitude and orbit during long transfers (for example to get to GEO or interplanetary orbits)	Propulsion involved Thrust parasitic torques to be controlled Typically autonomous
Mission Mode	• Fulfill particular targets of the mission	Various

Different ADCS modes may rely on different sensors/actuators, navigation, guidance and control
The design shall design the control modes, as well as **transitions** to switch from a control mode to another

1) Transitions Between Modes

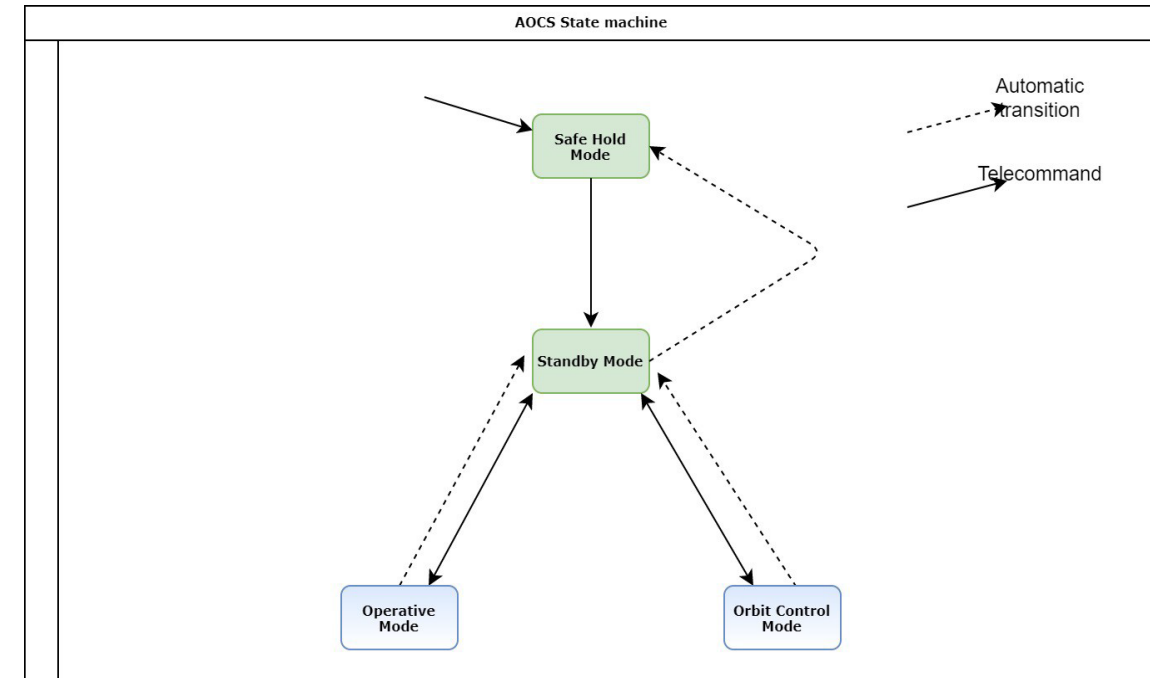
Control mode transitions need to be interfaced with system mode transitions.

The transitions may be *autonomous*, *automatic*, or *commanded* from ground.

- *Autonomous* transitions are typically event driven, commanded by internal logic (e.g. FDIR) and they shall never pose the spacecraft in danger
- *Automatic* transitions are typically time driven, executed after a task is completed, or when a schedule is on-going.

Good control mode architecture and transitions can be designed exploiting **Finite State Machines (FSM)** which are **mathematical models of abstract system behaviors**:

- FSM can be in exactly one of a finite number of states at any given time
- FSM change from one state to another (**transition**) in response to some inputs
- FSM are dynamics, discrete, finite



4) Evaluate maneuvers: Slew reqs and effects on ADCS actuators

Slewing reqs	Effects on s/c	Effects on ADCS
None or time unconstrained	s/c reorientation can take many hours	• Reaction wheels can be small • If magnetic torquers applicable, better than RW
LOW - 0.05-0.5°/s	Minimal	• Thrusters likely • Reaction wheels may be or may not adequate, depending on the s/c size
HIGH >0.5°/s	Structural impact on appendages Weight and cost increase	• CMG or thrusters required • Two level of thrusters: one for station-keeping + one for manoeuvres maybe needed

5) Quantify Disturbance Torques

External disturbances

(can be cyclic or constant)

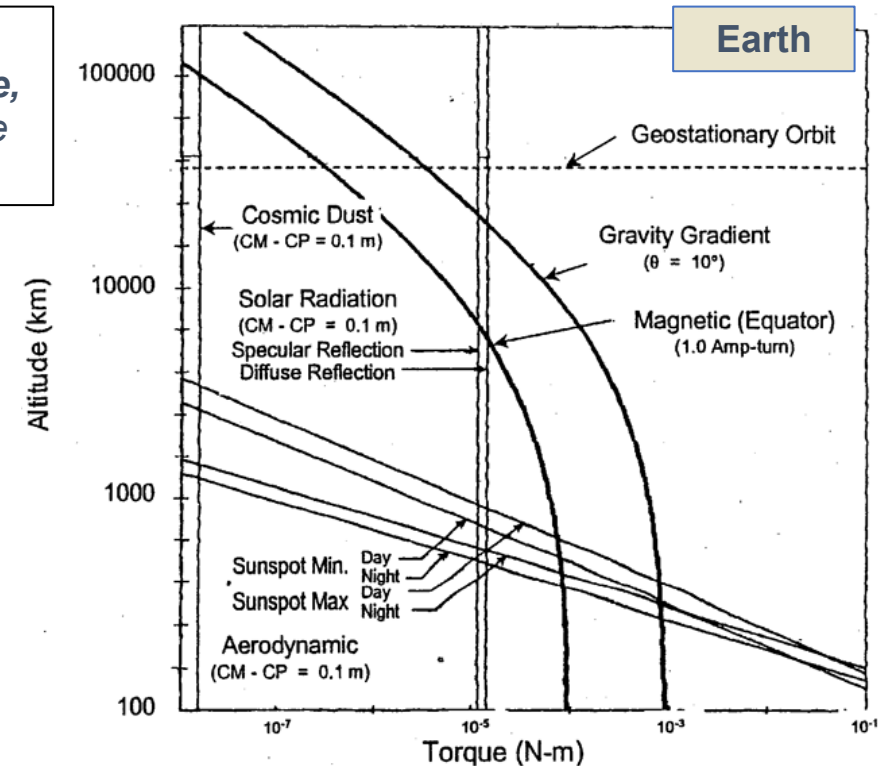
- Gravity gradient
- Atmospheric drag
- Solar radiation and pressure
- Magnetic moment

*Diagrams are strongly dependent on the **shape, design and inertia** of the spacecraft*

Internal disturbances

(constant or dynamic)

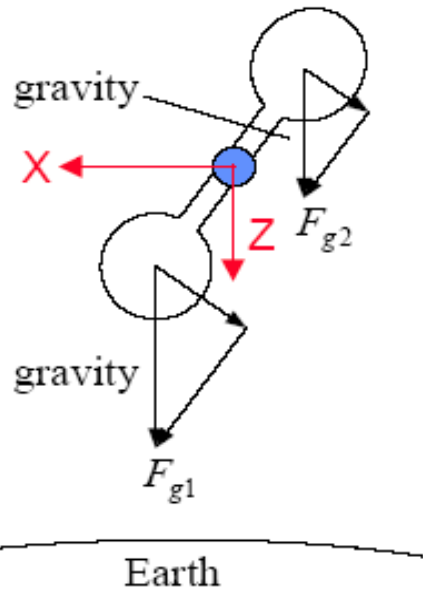
- Actuator misalignment (thruster, wheel, etc.)
- Sensor misalignment (gyro, magnetometer, etc.)
- Thrusters misalignment ($\rightarrow 0,1-0,5$ deg)
- Mismatch of thruster outputs: unbalanced torques when firing couples of thrusters ($\rightarrow -/+5\%$)
- Uncertainty in center of mass ($\rightarrow 1-3$ cm)
- Structural dynamics (such as arrays, booms, appendages, etc)
- Thermo-structural shocks (entering/leaving eclipse)
- Fluid slosh



5) Quantify Disturbance Torques

Gravity Gradient

- **Constant** disturbing torque for **Planet oriented**
- **Cyclic** disturbance torque for **inertially oriented**



- Gravitational force on mass m :

$$F_g = -\frac{\mu m}{R^2}$$

- A resulting torque occurs when

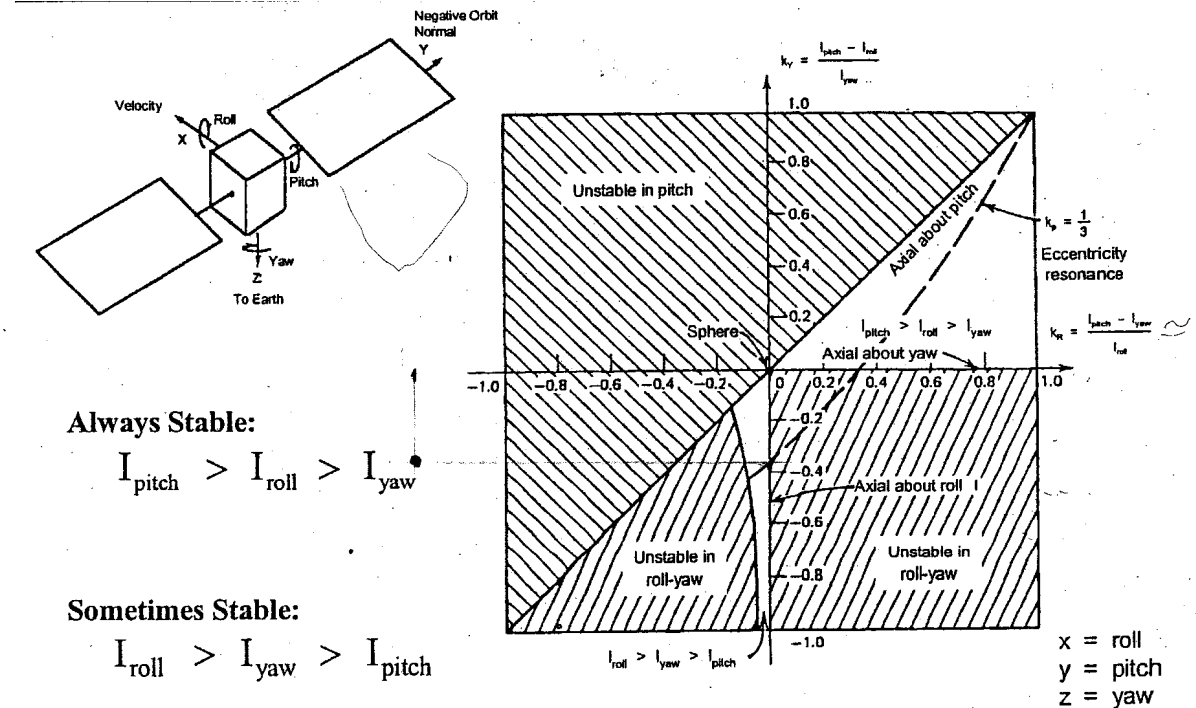
$$F_{g1} > F_{g2}$$

- In general, the gravitational torque can be expressed as

$$T_{gX} = \frac{3\mu}{2R^3} |I_Z - I_Y| \sin(2\theta_Y)$$

$$T_{gY} = \frac{3\mu}{2R^3} |I_Z - I_X| \sin(2\theta_X)$$

[X,Y,Z]= roll ,pitch, yaw axes



5) Quantify Disturbance Torques

Aerodynamic drag

- **Constant** disturbing torque for Planet oriented
- **Variable** disturbance torque for inertially oriented
- Different parts of a satellite have different drag coefficients
- This produces a net torque on the system that is

- Example: a low CG because of placing most components on the bottom of the satellite

$$T_d = F_d (c_p - c_g)$$

$$F_d = \frac{1}{2} (\rho C_d A V^2)$$

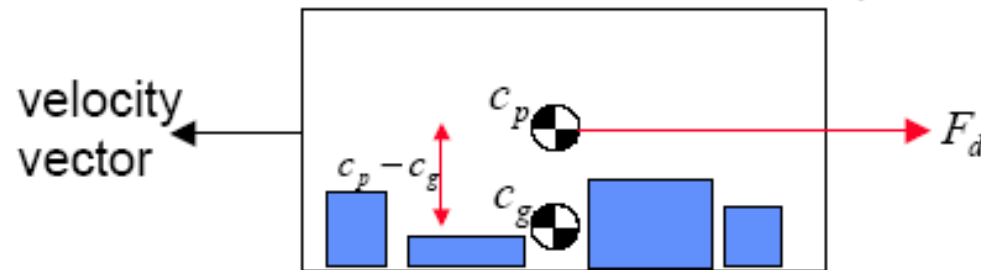
C_d = drag coefficient

c_p = center of aero pressure

A = surface area

V = velocity

ρ = atmospheric density



$C_d=2-2.5$

5) Quantify Disturbance Torques

Solar pressure

- **Cyclic** torque for **Planet oriented**
- **Constant** for solar oriented

$$\underline{F}_{\text{absorption}} = \frac{F_s}{c} A_s \cos \alpha \underline{u}_{\text{sun}}$$

$$\underline{F}_{\text{reflection}} = 2 \frac{F_s}{c} A_s \cos \alpha \underline{u}_{\text{normal,surf}}$$

$$\underline{F}_{\text{diffusion}} = \frac{F_s}{c} A_s \cos \alpha \underline{u}_{\text{sun}} + \frac{2}{3} \frac{F_s}{c} A_s \cos \alpha \underline{u}_{\text{normal,surf}}$$

Planet	Solar pressures, N/m ²
Mercury	3.05×10^{-5}
Venus	8.77×10^{-6}
Earth	4.59×10^{-6}
Mars	2.0×10^{-6}
Jupiter	1.70×10^{-7}
Saturn	5.10×10^{-8}
Uranus	1.24×10^{-8}
Neptune	5×10^{-9}
Pluto	3×10^{-9}

$$M_s = F(c_{ps} - c_g)(1 + q)$$

- α Solar Aspect Angle
 - F_s solar constant at spacecraft distance [W/m²]
 - c speed of light
 - c_{ps} solar pressure centre
 - c_g centre of gravity
 - q reflectance factor $\in [0 \ 1]$
- Ex. SA $\rightarrow$ $q=0.3$; general surface $q=0.5$

5) Quantify Disturbance Torques

Magnetic Field

- Cyclic

$$\underline{M}_m = \underline{D} \wedge \underline{B}$$

$$\underline{D} = N \cdot I \cdot A \underline{u}_{n,coil}$$

- $\underline{D}$ = residual dipole of the spacecraft (Am^2) (obtained from tests)
- N = # coils
- I = current in the coil (A)
- A = coil area (m^2)
- $\underline{B}$ = planet's magnetic field.

Planetary magnetic field usually developed in harmonic series as the gravity field but it can be approximated as follows:

$$B \approx \frac{B_0 R_0^3}{r^3} (3 \sin^2 L + 1)^{1/2}$$

- B_0 see-level magnetic field strength (T)
- R_0 Planet radius
- L magnetic latitude
- r orbital position vector magnitude

$$B_{\oplus,polar} \approx 2 \frac{B_0 R_0^3}{r^3}$$
$$B_{\oplus,equatorial} \approx \frac{B_0 R_0^3}{r^3}$$

Planet	Magnetic field strength, T	At radius, km
Mercury	3.5×10^{-7}	2,440
Venus	$< 3 \times 10^{-8}$	6,051
Earth	3.1×10^{-5}	6,378
Mars	5×10^{-8}	3,397
Jupiter	4.3×10^{-4}	71,372
Saturn	2.1×10^{-5}	60,330
Uranus	2.28×10^{-5}	25,600
Neptune	1.33×10^{-5}	24,765
Pluto	??	2,500

6) Select ADCS architecture

Method	Notes
Zero momentum	3 wheels: no constraint in pointing and maneuverability
	thrusters: no constraint in pointing and maneuverability at high rates possible
Gravity Gradient+ Momentum bias	Effective below 1000 km orbits (for Earth) • Roll and pitch axes can be controlled • yaw axis is stabilized by means of a momentum wheel • $I_{pitch} > I_{roll} > I_{yaw}$ always stable; $I_{roll} > I_{yaw} > I_{pitch}$ sometime stable
Momentum bias	A <u>momentum wheel</u> spins at nearly constant high speed • Local vertical pointing or inertial targets • It provides inertial stiffness in two axes • control of wheel speed provides control in the third axis.

6) Select ADCS architecture

Method	Notes
Spin stabilized	The s/c is stabilized along an axis by keeping an angular velocity around it. • With no disturbance the angular momentum keeps constant. • Perpendicular disturbances make the rotational axis to precess • Inertially fixed any direction • Repointing slow with torquers, faster with thrusters • Parallel disturbances change the angular momentum modulus. Translational manoeuvres may occur only along the spinning axis.
Dual spin stabilized	The major mass is spun while a platform with p/l or antenna is de-spun. • Limited only by the articulation on the despun platform

6) Select ADCS architecture

Method	Pros	Contrs/limitations
Zero momentum	• unlimited pointing capability in any direction • best pointing accuracy limited by sensors performances: pointing accuracy 0.0001 to 1deg (from wheels to thrusters) • <u>Large power requirements</u> admitted	• HW complex • Heavy • power consuming • <u>failure prone</u> • Expensive • active thrust vector control needed • <u>tanks require μ-g control.</u>
Gravity Gradient+momentum bias	• Good for modest pointing accuracy • Lifetime limited by the wheel bearings • Cheap • Robust • Limited power consumption	• Accuracy ± 5 deg • effective below a given altitude (e.g. 1000 km on Earth)
Momentum bias	• Good for long life missions • Cheap • <u>Good pointing accuracy in one axis</u> (0, 01 deg - pitch)	• poor accuracy in the not-wheel axes (1 deg - roll/yaw)

6) Select ADCS architecture

Method	Pros	Contrs/limitations
Spin stabilized	• good for scanning instruments • simple • cheap • useful during thrusters burning • Useful for propellant control	• low pointing accuracy, 0.1-1 deg on thwo axes • tight control of the moments of inertia required • the moment of inertia around the spinning axis must be the greatest not to reverse the vehicle • solar cells allocation may only be body mounted; low efficiency • slew rates must be $<0.5\text{deg/s}$ • body pointing of p/l or antenna is forbidden • Poor manoeuvrability
Dual spin stabilized	• good for scanning instruments and pointing capabilities • useful for propellant control • useful during thrusters burning	• Expensive • Complex • failure prone • attention must be paid in the distribution of inertial properties between the two s/c sections

ADCS pointing reqs versus architecture selection

<i>Pointing requirements</i>	<i>Type of architecture</i>
Control during kick stage firing	Spin stabilization preferred
Coarse control ($>10^\circ$)	Spin stabilization or passive control with gravity gradient
Low accuracy pointing ($>0.1^\circ$)	Dual spin or 3-axis control
High accuracy pointing ($<0.1^\circ$)	3-axis control
Low power requirement($<1\text{kW}$)	Oriented planar array or cylindrical spinning array
High power requirement($>1\text{kW}$)	Oriented planar array
Multiple pointing requirements	3-axis control
Attitude slewing requirement	Dual spin with articulation mechanism or 3 axis with wheels

7) Preliminary sizing

Inertial Features preliminary computation

Characteristic	Estimate	Range
Volume (m ³)	$V = 0.01 M$	0.005 to 0.05
Linear Dimension (m)	$s = 0.25 M^{1/3}$	0.15 to 0.30
Body Area (m ²)	$A_b = s^2$	—
Moment of Inertia (kg·m ²)	$I = 0.01 M^{5/3}$	—

7) ADCS HW sizing : Sensors

Sun sensors (0,005-3 deg)

- provide Sun direction vector (**NOT complete attitude state**).
- Not available during eclipses.
- Initialization not needed.
- Require Sun's ephemeris for absolute attitude reconstruction.

Star sensors (0,0003-0,01 deg)

- provide two- or three-axes attitude state.
- Require initialization (~seconds).
- Not suitable for large rates
 - **Star scanners:** used on spinning spacecraft. Light from different stars passes through multiple slits in the scanner's field of view.
 - **Star trackers:** used on three-axis stabilized spacecraft. Track (one or) more stars to derive (two-or) three-axis attitude information. Today, they are commonly CCD creating images of the star field, to be compared with a star catalogue.

Horizon (Earth) sensors (0,05-1 deg)

- provide Earth-relative information directly for Earth-pointing spacecraft
- Require accurate orbital data for absolute attitude reconstruction.

Any two of these vector measurements provide four independent measurements, allowing to determine the orientation of the vehicle in space

7) ADCS HW sizing : Sensors

Magnetometers (0,5-3 deg)

- provide magnetic field direction vector and magnitude
- Require accurate magnetic model for absolute attitude reconstruction.
- Needed for magnetic torquers usage.

Gyroscopes

- provide angular rates, and attitude evolution.
- Need for an external attitude reference to have absolute attitude states.
- Can be used for attitude propagation during loss of external attitude reference.
- On-board calibration of gyroscopes (e.g. bias estimation) for accurate attitude propagation.
- Bias drift stability and bias drift are the most relevant performance for gyro selection.

7) ADCS HW sizing : Sensors

GNSS

- provide orbit position (may be needed to feed internal models used for attitude reconstruction (e.g. magnetic field models, Earth absolute position ...)).
- Can be used for attitude reconstruction using **differential GNSS** measurements (e.g. 0.25-0.5 deg accuracy for 1m differential antenna baseline).

Accelerometers

- provide accelerations that can be used for orbit propagation and orbit determination filtering when used together with GNSS (e.g. GPS + Accelerometers in **INS applications**).
- **Not sensing the external gravitational acceleration** (need for a gravitational model – e.g. Kep, J2, Spherical harmonics...) for orbit propagation.
- Very useful for correct orbit maneuvers actuation and for accurate on-board orbit determination

IMU (inertial measurement unit)

- set of 3 gyros and 3 accelerometers for complete attitude and position sensing.

7) ADCS HW sizing : Sensors data

Sensor	Typical Performance Range	Mass (kg)	Power (W)
Gyroscopes	Drift Rate = 0.003 deg/hr to 1 deg/hr Drift rate stability varies widely	< 0.1 to 15	< 1 to 200
Sun Sensors	Accuracy = 0.005 deg to 3 deg	0.1 to 2	0 to 3
Star Sensors (Scanners & Cameras)	Accuracy = 1 arcsecond to 1 arcminute = 0.0003 deg to 0.01 deg	2 to 5	5 to 20
Horizon Sensors	Accuracy:		
- Scanner/Pipper	0.05 deg to 1 deg (0.1 deg is best for LEO)	1 to 4	5 to 10
- Fixed Head (Static)	< 0.1 deg to 0.25 deg	0.5 to 3.5	0.3 to 5
Magnetometer	Accuracy = 0.5 deg to 3 deg	0.3 to 1.2	< 1

7) ADCS HW sizing : accuracy effects on selection

Req. accuracy	Effects on s/c	Effects on ADCS
>5°	• GG can be applied • Cost savings	without attitude determination • No sensor required for GG • Boom motor, GG damper With Attitude determination • Sun sensor and magnetometers at >2 deg
1-5°	• GG unfeasible • If inertial pointing ok, spin acceptable • 3 axis well suited	• Sun and horizon sensors suggested especially for spinning s/c • RCS can be selected with thrusters for desaturations • Magnetic system for adequate orbit • Thrusters and dumper for spinners
0.1-1°	• 3axis well suited • Dual spin feasible	• Star sensors and gyros for precise reference • Reaction wheels+thrusters for desaturation
<0.1°	• 3axis necessary • May ask for isolated p/l platform	• More precise star sensors and gyros for precise reference • Control laws more complex

7) Select and size ADCS HW: Actuators

Reaction Wheels

Torque motors with high inertia rotors: nominal zero momentum; they can spin in either directions and provide one axis of control each; torque is applied by changing the rotor speed; control to counteract disturbances and to provide slew manoeuvres

Momentum Wheels

Reaction wheels with nominal spin different from zero. They provide a nearly constant momentum. Stiffness is provided on two axes and the motor torque control the third axis pointing.

CMG

They generate torque by using torque motors to change the direction of the wheel's spin axis. They rotate at high speed with small inertia. CMG are moveable angular momentum vectors that can be controlled by means of motors in the gimbals. Applied for large torques or fast manoeuvres; massive but more efficient than RW

Thrusters	Actuator	Typical Performance Range	Mass (kg)	Power (W)
	Thrusters	Thrusters produce force; multiply by moment arm for torque	Mass and power vary (See Ch. 18 for details on propulsion systems)	
	Hot Gas (Hydrazine)	0.5 to 9000 N		
	Cold Gas	< 5 N		
Magnetic torquers	Reaction and Momentum Wheels	Maximum torques: 0.01 to 1 N·m Practical momentum storage capacity: 0.4 to 3000 N·m·s	2 to 20	Varies with speed: 10 to 100
	Control Moment Gyros (CMG)	Max. torques: 25 to 500 N·m	> 10	90 to 150
	Magnetic Torquers	1 to 4,000 A·m ²	0.4 to 50	0.6 to 16

Actuators must have sufficient torque authority to counteract disturbances.
Control Authority: Control Torque - disturbance torque
ex. CT=2DT → 100% C.A. margin

7) Select and size ADCS HW: Actuators - RW

										example
Characteristic	Mini-wheel	HR 0610	HR 12	HR 14	HR 16	HR 4820	HR 2010	HR 2020	HR 2030	HR 4520
Angular momentum, N-m-s	0.2 to 1.0	4 to 12	12 to 50	20 to 75	75 to 150	65	33.2 to 68.4	27	19.5 to 45.6	60.75
Output torque, N-m	>0.028	0.07 to 5	0.1 to 0.2	0.1 to 0.2	0.1 to 0.2	0.14	0.1	0.13	0.21	0.135
Wheel rpm, $\pm$	9000	6000	6000	6000	6000	6000	6000	6500	6000	5400
Power, W ^b	>6	<15	22	22	22	20	17	35	20	35
Bus voltage, dc	12 to 34	14 to 35	23 to 57	23 to 57	23 to 57	22 to 36	27 to 44	70	27.7 to 31.3	51
Mass, kg	1.3	3.6 to 5.0	7.0	8.5	12	10.2	9.2 to 10.9	7.9	8.9 to 11.2	11.1
Integral electronics	Y	Y	Y	Y	Y	Y	N	Y	Y	Y
Diameter, mm ^c	108	267	316	368	418	405	406	300	305	406
Height, mm	54	12.0	159	159	178	214	235	172	191	215
Op temperature, ^d										
Low	-25	-15	-30	-30	-30	-15	-15	-13	-15	-24
High	+60	+60	70	+70	+70	+71	+70	+75	+80	+61

<https://aerospace.honeywell.com/us/en/products-and-services/product/hardware-and-systems/space#5>

<https://aerospace.honeywell.com/us/en/products-and-services/product/hardware-and-systems/space/small-satellite-specific-bus-products/hr04-hc7-hc9-reaction-wheel-assemblies>

<https://www.collinsaerospace.com/what-we-do/industries/space/space-wheels>

<https://www.rocketlabusa.com/space-systems/satellite-components/reaction-wheels/>

6) ADCS Manoeuvring sizing

Assumptions

- Constant angular acceleration
- Single axis maneuvering



$$\begin{cases} \vartheta = \frac{1}{2} \alpha t_m^2 \\ \omega = \alpha t_m \\ \alpha = \frac{M}{I_v} \\ H = I_v \omega \end{cases} \Rightarrow H = M t_m$$

α vehicle angular acceleration (s^{-2})
 ω vehicle angular velocity (s^{-1})
 t_m actuator switch on time
 M control torque
 I_v vehicle moment of inertia
 H vehicle angular momentum

6) ADCS Manoeuvring sizing with thrusters

Zero momentum architecture

Thrusters → obtainable torque M: $M = nFL$

n = # thrusters

t_b = burning time

L = lever arm for thrusters

t_b = burning time

H = angular momentum

$$\left\{ \begin{array}{l} \alpha = \frac{nFL}{I_v} \\ \mathcal{G} = \frac{1}{2} \frac{nFL}{I_v} t_b^2 \\ H = nFLt_b \\ I_{sp} = \frac{nFt_b}{g_0 m_{fuel}} \end{array} \right. \Rightarrow m_{fuel} = \frac{nFt_b}{I_{sp}} = \frac{H}{L \cdot I_{sp}}$$

$I_{sp} \uparrow \uparrow$
 $L \uparrow \uparrow$

6) ADCS slew manoeuvring sizing with thrusters

Zero momentum architecture

Manoeuvre to get a desired θ_m

→ Manoeuvre sequence:

- 1) Constant acceleration
- 2) Coasting (optional)
- 3) Constant braking

$$\vartheta_m = \vartheta_{ac} + \vartheta_{coast} + \vartheta_{brake}$$

$$\vartheta_{coast} = \omega t_{coast}$$

$$\omega_{coast} = \frac{nFL}{I_v} t_b$$

$$\vartheta_{ac} = \frac{nFL t_b^2}{2I_v}$$

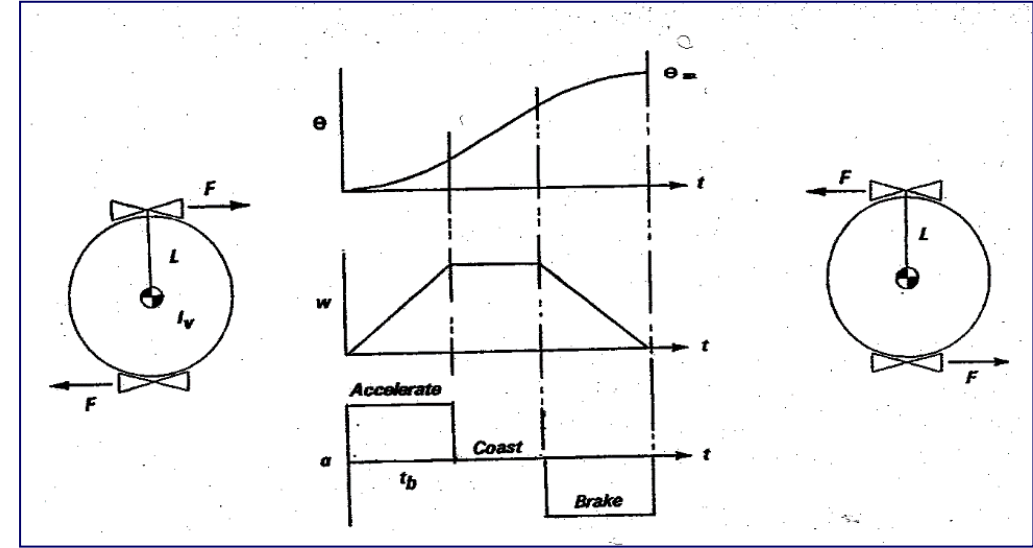
$$\Rightarrow \vartheta_m = \frac{nFL t_b^2}{I_v} + \frac{nFL}{I_v} t_b t_{coast}$$

Manoeuvre with desired θ_m

- 1) acceleration
- 3) braking

$$t_m = 2t_b = 2\sqrt{\frac{\vartheta_m I_v}{nFL}}$$

$$m_p = 2 \frac{nF t_b}{I_{sp} g_0}$$



6) ADCS slew manoeuvring sizing with RW

Manoeuvre to get a desired θ_m

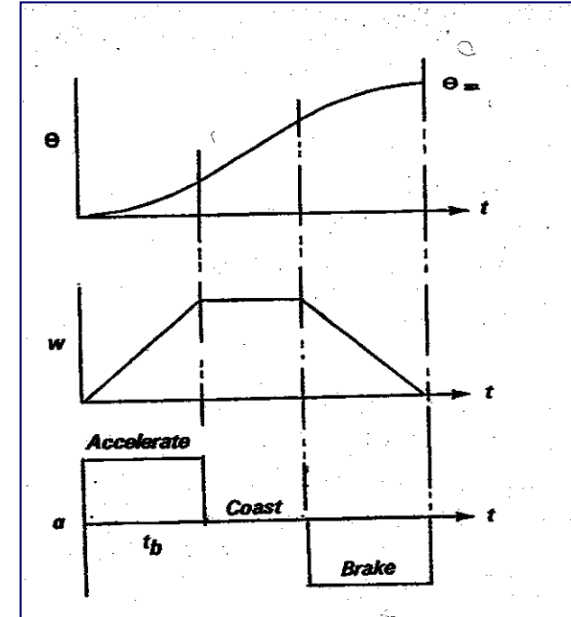
1) acceleration; 2) coasting=0; 3) braking

$$\vartheta_m = \vartheta_{ac} + \vartheta_{brake}$$

$$\vartheta_m = \frac{Mt_m^2}{4I_v} \Rightarrow \vartheta_m = \frac{\alpha_w I_w t_m^2}{4I_v}$$

$$M_{vehicle} = \alpha_w I_w$$

Note: the net change in wheel speed is zero



Wheels may need to be de-saturated for unbalanced angular momentum:

$$I_{desat} = nFt = \frac{I_w \omega_w}{L}$$

$$I_w \omega_w = H = Mt = nFLt \Rightarrow$$

$$m_p = \frac{I_w \omega_w}{LI_{sp}g_0}$$

$$t_{desat} = \frac{I_w \omega_w}{nFL}$$

6) ADCS manoeuvring limit cycle

No external torque

The s/c is swung between preset angular limits.

Note:

The correction pulse should be kept as smaller as possible as it must be reversed

$$Period_w = 2t_b$$

$$\vartheta_m = \frac{nFL}{I_v} \left(\frac{P_w^2}{4} + \frac{P_w t_{coast}}{2} \right)$$

$$\vartheta_{limit} = \frac{nFL}{4I_v} P_w t_{coast} \Rightarrow t_{coast} = \frac{4I_v}{nFLP_w} \vartheta_{limit}$$

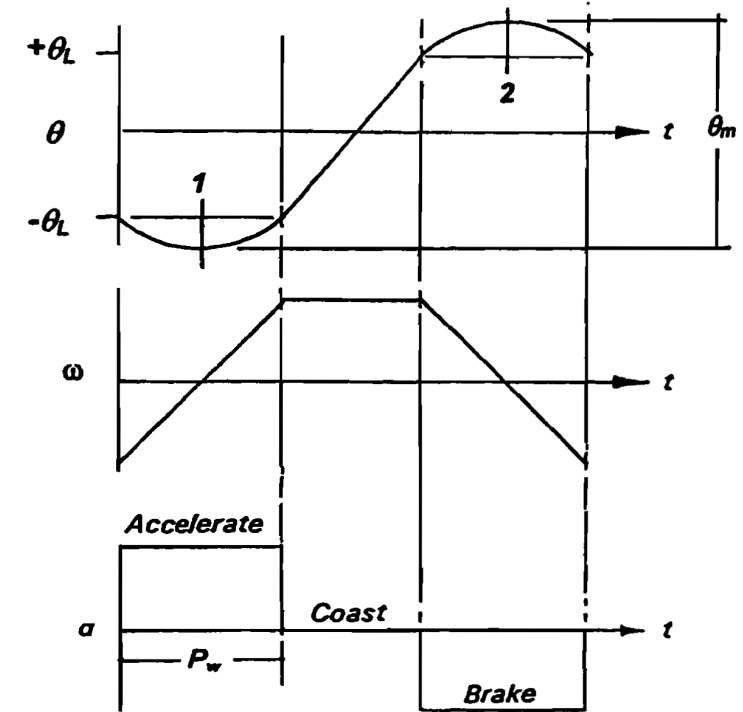
$$m_{fuel,cycle} = \frac{2nFP_w}{I_{sp}}$$

Mass per cycle. Short pulses and high Isp preferred

$$t_{cy} = \frac{8I_v \vartheta_{lim}}{nLI_{min}} + 2P_w \rightarrow P_w \ll \ll \ll$$

$$\frac{m_p}{t_{cy}} = \frac{n^2 I_{min}^2 L}{4I_{sp} I_v \vartheta_{lim}} = \dot{m}_p$$

Propulsion system type	Min thrust, N	Min impulse bit, N-s	Pulsing I_{sp} , s
Cold gas-helium	0.05	0.0005	80
Cold gas-nitrogen	0.05	0.0005	50
Monopropellant-N ₂ H ₄	0.5	0.005	120
Biopropellant-N ₂ O ₄ /MMH	8	0.025	120



6) ADCS manoeuvring

one side limit cycle

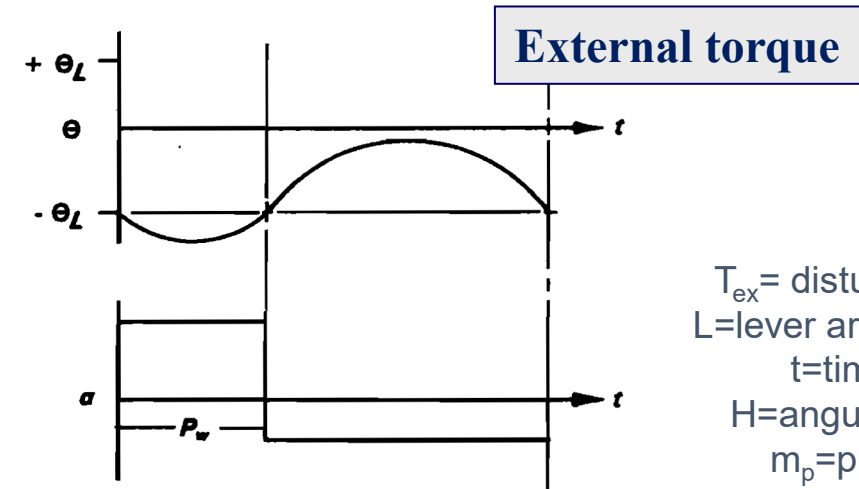
The s/c is disturbed by external torque

Note:

The correction pulse should be kept as smaller as possible

$$H = T_{ex} \cdot t$$

$$m_p = \frac{H}{I_{sp}L}$$



T_{ex} = disturbance torque
 L = lever arm for thrusters
 t = time to boundary
 H = angular momentum
 m_p = propellant mass

Spinning re-orienting

$$\phi = \frac{H_{control}}{H_{vehicle}} = \frac{nFLt}{I_{v,spin}\omega}$$

